

Vaidehi Gohil

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EDUCATION

Northeastern University, Boston, MA

May 2027

Master of Science in **Electrical and Computer Engineering** (Concentration: Computer Systems and Software)

Courses: Deep Learning Embedded Systems, Wireless Sensors, High Performance Computing

Graduate Research Assistant at [Mon\(IOT\)or Lab](#) contributing to the SPHERE IoT testbed project, an open infrastructure for IoT security and privacy research with 100+ devices.

Vidyavardhi's College of Engineering and Technology, Mumbai, India

June 2024

Bachelor of Technology in Electronics and Telecommunication Engineering

Relevant Courses: Microcontrollers, Artificial Neural Networks, Linear Integrated Circuits, Digital Signal Design
(Patent Published by Govt. of India, No. 202521062488)

TECHNICAL SKILLS

Embedded Systems & Microcontrollers:	Raspberry Pi, Nvidia Jetson Orin Nano, RPi Pico & Zero W, ESP-WROOM32, Arduino Boards, Nordic nRF7002 Board, HMI, MSP430
Communication Protocols & Electronics:	I ² C, SPI, UART, RS-485, Modbus, OPC UA, BLE, MOSFET & ICs, Power-Path Switching (BMS), IMU, Sensor Integration
Software, ML & System Analytics:	Linux (Shell, systemd), C/C++, Python, ROS2, Git, CNN, Machine Learning, System Validation, Sensor Data Analytics, Windows
Hardware:	3D Printing, Laser Cutting, Soldering, Woodworking (Table Saw, Grinding)

WORK EXPERIENCE

Faclon Labs, Mumbai, India

November 2024 – August 2025

IoT Application Engineer

- Built fault-tolerant Python Modbus (RTU/TCP) poller and TCP telemetry services on OpenWrt with timeout validation, retry/backoff, and watchdog auto-recovery, deployed across 5+ remote sites to maintain continuous data collection with minimal manual intervention
- Integrated heterogeneous devices (I²C, SPI, UART, RS-485 & BLE); defined register maps and polling schedule; standardized outputs as structured JSON via MQTT/HTTP for dashboards and faster field diagnostics
- Validated in-house BMS power-path switching between battery and external supply, which was done using three MOSFET switching stage, ensuring expected state transitions and functionality during system bring-up

PROJECTS

Telehealth: Connectivity to Wellness, Mumbai, India

August 2023 – January 2024

- Designed and prototyped a telehealth kiosk for rural settings, integrating ECG, SpO₂, and temperature sensing for basic checkups and remote consultation
- Implemented a two-tier architecture with ESP-WROOM for sensor calibration, and has a Raspberry Pi with a web dashboard for real-time visualization and remote video consults
- Developed sensor drivers and data aggregation code to produce stable readings and reliable delivery to the dashboard, documented system design with calibration and usage procedures for future deployment

Warehouse Drone Autonomy Stack, Boston, MA

November 2025 – Present

- Building an indoor GPS-denied navigation system on Jetson Nano (Linux environment) using ROS2 and PX4, integrating IMU, LiDAR and Stereo Camera for state estimation, odometry, and localization in a warehouse